

Status

Feature Status Ledger

Revision: 15 **Last modified:** 2026-06-16T14:05:00Z **Authority:** constitution §11.4.153 (per-feature Status + video-recording confirmation), composes §11.4.44 (revision header), §11.4.45 (integration-status-doc + closed status vocabulary), §11.4.56 (Status_Summary two-audience companion), §11.4.65 (HTML+PDF export) + §11.4.153 DOCX, §11.4.107 (no frozen-frame video proof), §11.4.6 (no-guessing — every row reconciled against real code). **Scope:** Full feature surface of the ytdlp Podman/Docker orchestration around yt-dlp — every service, client surface, CLI script, landing route, dashboard component, test/challenge suite, and every PLANNED-but-unbuilt feature. **Maintainer:** project conductor.

How to read this ledger

- **Implementation** — Implemented (code present + wired), Partial (code present, incomplete/edge-gaps), Planned (not yet built).
- **Validation/Verification** — closed status vocabulary per §11.4.45: PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED. Because no clean-baseline full-suite run with captured evidence was executed in the session that produced this ledger, verdicts that would require live captured runtime evidence are recorded PENDING_FORENSICS (per §11.4.6 — an unproven PASS is forbidden; the code is present and test files exist, but a green-with-captured-evidence run is not yet on record here). Planned rows carry SKIP (no feature to validate yet).
- **Video-Confirmation** — per §11.4.153 every user-visible confirmed claim MUST be backed by a recorded real-use capture. The strong-model analysis path is the agent's OWN native multimodal read (Claude Opus 4.8 via the Read tool — see docs/research/vision-path/FINDINGS.md); local CPU models (moondream) are NOT used (too slow + hallucinate). **NINE render/transcript captures are confirmed — dashboard / (download-form + navbar), dashboard /queue, dashboard /history, dashboard /cookies, landing 'Боба' UI, MeTube UI, postprocess status API, ./status CLI, ./download CLI (together confirming the dashboard's five components + the other surfaces) PLUS a landing read-only-API transcript (ytdlp---landing-api---*) confirming seven §4 routes (/ , /health, /api/cookie-status, /api/profile-status, /api/aborted-history GET, /logo.png, /favicon.ico) — AND ONE end-to-end FLOW is confirmed (download→webready transcode, via advancing live-pipeline counter + ffmpeg-validated just-produced artifact). Every OTHER row remains PENDING — not yet recorded until its window-scoped, ytdlp---prefixed real-use capture (§11.4.154/.155) is recorded + analyzed. The four running §1**

services (metube, landing, dashboard, yt-dlp-cli) are also marked service-serving-confirmed, each citing its surface capture(s). The remaining PENDING rows are dominated by **mutating** routes/flows (POST/DELETE cookie-upload, delete-download, aborted-history write) which are NOT auto-captured because they would alter operator state (§9.2/§11.4.122), the MP3-derivation flow (no audio artifact yet), the not-running services (VPN profile — operator .ovpn; watchtower — docker profile), and the state-changing root CLI scripts (./start/./stop/./restart/./cleanup/./init — running them would destabilize the live stack).

HONEST FACT (§11.4.6 / §11.4.153): As of this revision the real PASSes span TEN distinct ytdlp----prefixed captures. NINE are scope=render/transcript: seven image renders analyzed by Claude Opus 4.8 native multimodal — dashboard / (ytdlp---dashboard---20260615T221723Z.png, download-form + navbar), dashboard /queue (ytdlp---dashboard-queue---20260616T083018Z.png, empty state), dashboard /history (ytdlp---dashboard-history---20260616T083023Z.png, content state, 46 items), dashboard /cookies (ytdlp---dashboard-cookies---20260616T083029Z.png, content state, per-platform), landing 'Боџа' UI (ytdlp---landing---20260616T074850Z.png), MeTube UI (ytdlp---metube---20260616T075608Z.png), postprocess status API (ytdlp---postprocess---20260616T075648Z.png) — plus three transcripts: ./status (ytdlp---status---20260616T075901Z.txt), ./download (ytdlp---download-cli---20260616T080936Z.txt), and the landing read-only-API sweep (ytdlp---landing-api---20260616T083612Z.txt, seven §4 routes each HTTP 200/204 with real bodies). ONE is scope=FLOW: the download→webready transcode (ytdlp---webready-flow---20260616T080633Z.txt), confirmed by the live pipeline's done counter advancing 25→26 across two real status snapshots plus ffmpeg-validation of the just-produced 1920×1080 h264 +faststart artifact (the postprocessor is a backend with no UI, so the §11.4.69/.107 sink-side evidence is the validated artifact, not a screen recording). **Still NOT flow-confirmed:** MP3-derivation and cookie-upload FLOWS (the latter operator-gated — needs fresh operator cookies; uploading test cookies would overwrite real operator state, §9.2/§11.4.122; the /cookies render shows YouTube + X cookies expired 2d ago, corroborating the gate). Every other cell honestly reads PENDING — not yet recorded. No row over-claims.

OPERATOR-BLOCKED rows (top per §11.4.45)

No feature in this project is currently OPERATOR-BLOCKED. (The VPN profile requires operator-supplied .ovpn credentials and vpn-auth.txt, but the VPN *code path* is implemented and testable in compose-config form without live credentials, so it is classified PENDING_FORENSICS, not OPERATOR-BLOCKED.)

Rev 10 — captured retest + fresh re-records (§11.4.40 safe subset, parallel subagents §11.4.103): the media_postprocessor code-under-test is now PASS with captured evidence — **69/69 pytest** (incl. the new test_skip_rule_regression.py §11.4.135 guard) + **smoke 22/0 + contract 18/0 GREEN** against the live stack, raw output at qa-results/

retest-20260616T091806Z/. All four UI surfaces were re-recorded fresh + re-read natively (4/4 PASS): ytdlp---
dashboard---20260616T091827Z.png, ytdlp---
landing---20260616T091840Z.png, ytdlp---
metube---20260616T091849Z.png, ytdlp---
postprocess---20260616T091900Z.png. The container-lifecycle suites (scenario/integration/full-suite) were deliberately NOT run — they would tear down the released ytdlp-1.4.0 stack — so rows depending on them stay PENDING_FORENSICS. Issue triage (systematic-debugging §11.4.102): the done-counter-frozen + the 11 failed jobs were both proven NOT bugs (a long live transcode; pre-fix stale intermediate-file entries) — see docs/research/vision-path/ + commit history.

Rev 11 — Round-2 parallel subagents (189 tests green this session): added **Angular dashboard unit 63/63 SUCCESS** (download-form/queue/history/cookies components + metube.service ×2 → those §5 Validation cells now PASS) and **pure-shell unit 17/17 PASS** (tests/test-unit.sh) — evidence qa-results/retest-20260616-round2.md. Cumulative captured-green this session = 69 pytest + 22 smoke + 18 contract + 63 Angular + 17 shell = **189 tests, 0 real regressions**. Vision (§11.4.150) was EMPIRICALLY VALIDATED on this host — mlx-vlm + Qwen2.5-VL-3B (Metal) does ~20 s/frame with a grounded UI description (good-not-flawless; native-multimodal stays primary) — see docs/research/vision-path/CPU_VISION_RESEARCH.md.

Rev 13 — PRIMARY DEPLOYMENT MOVED TO nezha.local (§11.4.76 distributed booting; local Mac stack stopped): the full no-vpn System was distributed to nezha (amd64, Podman 5.7.1) via scripts/deploy-remote.sh + deploy/remote-hosts.env.example, and HEAVY-TESTED against real running production services — **captured evidence in qa-results/nezha-heavytest-20260616/** (+ SUMMARY.md): run-tests scenario **17/0** + integration **123/1** pre-fix → **124/0 clean** post-fix (the 1 fail = test_metube_api_vk VK-playlist-URL allowlist, env — fixed; the post-fix full-integration re-run 2b-integration-postfix.log is "All tests passed!") + error **24/0**, container-restart PASS, chaos **27/0/1**skip(DNS), memory-limits PASS, smoke **22/0**, contract **18/0**, integration-realhttp **22/0**, 7/7 challenges (incl. download_completes → a real **111 MB webready** file on disk), postprocessor pytest **46p/23skip/0fail**, dashboard 34/1 (env: stale-cookie-size, re-seeded), media-services **16/1** (same VK env fail, fixed), aborted-history 9/0, bulk 9/0. **Zero product bugs**; 6 deploy/test issues root-caused + fixed (§11.4.102, commits 5d15a64/f10c727/acbed76/b96dc8f). The 4 user surfaces were **re-recorded on nezha + re-read natively (4/4 PASS)**: ytdlp---nezha-dashboard---20260616T135945Z.png (full Add-Download UI + 16-platform grid), ytdlp---nezha-metube---20260616T140007Z.png (real fresh downloads "M5 Ultra Mac Studio Leaks..." 62.62 MB etc.), ytdlp---nezha-postprocess---20260616T140014Z.png ({"healthy":true,...,"done":7,"failed":0} — clean pipeline), ytdlp---nezha-landing---20260616T140001Z.png (the :8086→:9090 auto-redirect fired, as designed). The §1 services + §3 CLI scripts + §6 test/challenge suites are therefore now backed by captured green runs on a REAL Linux+Podman production host (the heavy container-lifecycle suites that could NOT run on the Mac).

1. SERVICES (docker-compose.yml)

Feature	Component	Category	Implementation	Wiring
VPN tunnel (dperson/openvpn-client, NET_ADMIN, /dev/net/tun, port 3130, healthcheck ping 8.8.8.8)	openvpn-yt-dlp	Networking / Service	Implemented (vpn profile)	Reachable: metube + yt-dlp its netns via network_mode service:openvpn-yt-dlp
Download queue/ history API + minimal UI (ghcr.io/ alexta69/ metube:latest, port 8088 direct / netns under VPN)	metube / metube-direct	Backend API / Service	Implemented	Reachable: dashboard nginx to :8081; \${DOWNLOAD_DIR} config volumes
Flask 'Боџа' cookie- auth onboarding + aborted-history ledger + proxy endpoints (landing/, ports 8087 vpn / 8086 no- vpn)	landing- vpn / landing-no- vpn	Web UI + API / Service	Implemented	Reachable: published port; p delete-download to MeTube aborted.json
Sleeping CLI container for ./ download (ghcr.io/ jim60105/yt- dlp:pot, while true; do sleep 3600; done, cookie copy at startup)	yt-dlp-cli / yt-dlp-cli- vpn	CLI backend / Service	Implemented	Reachable: ./download exe it; yt-dlp/config, cookie downloads volumes
Angular 17 + nginx dashboard (dashboard/, port 9090, /api/* proxied to MeTube + landing)	dashboard	Web UI / Service	Implemented (no-vpn profile only)	Reachable: nginx resolver proxy_pass from entryp

Feature	Component	Category	Implementation	Wiring
Image auto-update (containrrr/watchtower:latest, schedule 0 0 */3 * * *, cleanup)	watchtower	Infra / Service	Implemented (docker profile)	Reachable: docker.sock mount com.centurylinklabs.watchtower

2. CLIENT SURFACES

Feature	Component	Category	Implementation	Wiring	Tests Coverage
Dashboard web UI (download-form / queue / history / cookies / navbar, error-boundary, not-found)	dashboard	Web client	Implemented	Routed via app.routes.ts; calls /api/* through nginx	tests/test-dashboard e2e/tests/dashboard component *.spec.ts
Landing Flask 'Боба' UI (3-step cookie onboarding, drag-drop upload, freshness banner, services grid)	landing	Web client	Implemented	INDEX_TEMPLATE in landing/ app.py; auto- redirect to dashboard	tests/e2e/tests/ landing.spec.ts, t dashboard.sh
MeTube minimal UI (vendor original interface, dark theme)	metube	Web client	Implemented (vendor)	Served on :8088; linked as "Classic UI" from landing	tests/test-media-

Feature	Component	Category	Implementation	Wiring	Tests Coverage
./download CLI (host script execs yt-dlp in sleeping container)	download	CLI client	Implemented	./download → exec into yt- dlp-cli; runtime auto- detect	tests/test-media- challenges/script download_complete

3. ROOT CLI SCRIPTS

Feature	Component	Category	Implementation	Wiring	Tests Coverage
Init environment (create dirs, validate .env, generate yt- dlp.conf, write vpn- auth.txt chmod 600)	./init	CLI script	Implemented	Runtime auto-detect via lib/ container- runtime.sh	tests/test-unit.sh, t integration.sh, tests
Start services (reads USE_VPN, pulls images, brings up profile)	./start	CLI script	Implemented	compose up via detected runtime	tests/test-scenarios integration.sh
Start no-vpn (force no- vpn profile)	./start_no_vpn	CLI script	Implemented	compose -- profile no-vpn	tests/test-scenarios scripts/ no_vpn_profile_direc
Stop services (all profiles, clean pods/ networks)	./stop	CLI script	Implemented	compose down + cleanup	tests/test-scenarios

Feature	Component	Category	Implementation	Wiring	Tests Coverage
Status summary (service + HTTP health)	./status	CLI script	Implemented	queries running containers + HTTP endpoints	tests/test-integration
Download helper (yt-dlp via exec)	./download	CLI script	Implemented	exec into yt-dlp-cli	tests/test-media-server scripts/download_com
Check VPN (query ipinfo.io inside VPN container)	./check-vpn	CLI script	Implemented	exec into openvpn- yt-dlp	tests/test-vpn-smoke
Restart services	./restart	CLI script	Implemented	stop + start sequence	tests/test-scenarios
Cleanup containers/pods/networks	./cleanup	CLI script	Implemented	compose down + prune	tests/test-scenarios scripts/ container_restart_re
Release preparation (commit + push)	./prepare-release.sh	CLI script	Implemented	git operations	bash -n syntax gate (make
Auto-update setup (cron for Podman; Watchtower alt for Docker)	./setup-auto-update	CLI script	Implemented	writes cron entry / documents Watchtower	bash -n syntax gate
Update images (pull)	./update-images	CLI script	Implemented	compose pull via	bash -n syntax gate

Feature	Component	Category	Implementation	Wiring	Tests Coverage
latest before start)				detected runtime	

4. LANDING ROUTES (landing/app.py)

Feature	Component	Category	Implementation	Wiring
/ — Бoбa onboarding page (3-step cookie flow, session id, dynamic service URLs)	landing	HTTP route	Implemented	render_template_string(INDEX_TE
/app — proxy GET to MeTube backend	landing	HTTP proxy route	Implemented	requests.get(METUBE_URL) streamec
/api/upload-cookies (POST) — validate Netscape file (UTF-8 + recognised-domain gate) → forward to MeTube	landing	HTTP API route	Implemented	_validate_cookie_file + POST to \$ {METUBE_URL}/upload-cookies
/api/delete-cookies (POST) — remove / config/ cookies.txt	landing	HTTP API route	Implemented	os.remove
/api/cookie-status (GET) — has_cookies + age + per-platform breakdown + MeTube reachability	landing	HTTP API route	Implemented	_summarize_cookies_by_platform+ MeTube fallback
/api/aborted-	landing		Implemented	_read_aborted_history (lock-free re

Feature	Component	Category	Implementation	Wiring
history (GET) — return persisted aborted list		HTTP API route		
/api/aborted-history (POST) — append entry, flock-guarded, 60s idempotency	landing	HTTP API route	Implemented	<code>_aborted_history_lock + _write_aborted_history</code>
/api/aborted-history (DELETE) — remove specific urls or *	landing	HTTP API route	Implemented	locked read-modify-write
/api/profile-status (GET) — active compose profile + vpn_active + MeTube reachability	landing	HTTP API route	Implemented	reads <code>ACTIVE_PROFILE</code> env
/health (GET) — service + MeTube reachability + timestamp	landing	HTTP API route	Implemented	<code>requests.get("\${METUBE_URL}/hist</code>
/logo.png (GET) — serve branding asset	landing	HTTP static route	Implemented	<code>send_from_directory</code>
/favicon.ico (GET) — 204 stub	landing	HTTP static route	Implemented	returns <code>""</code> , 204
/api/delete-download (POST) — remove from	landing	HTTP API route	Implemented	<code>POST \${METUBE_URL}/delete + os.w</code> <code>os.remove</code> within <code>DOWNLOAD_DIR</code>

Feature	Component	Category	Implementation	Wiring
history + optional file delete (path-traversal guarded)				

5. DASHBOARD COMPONENTS (dashboard/src/app)

Feature	Component	Category	Implementation	Wiring
Download form (url, quality select, format select, folder, add via /api/add, inline tracker + error state, re-enable after failure)	download-form	Angular component	Implemented	MeTubeService.a
Queue (STATE_META: pending / preparing / downloading / postprocessing / finished / error / aborted; cancel-all; record-and-close aborted)	queue	Angular component	Implemented	getHistoryPolli recordAbortedIt
History (+ aborted-history merge via abortedToDownloadInfo, retry, delete-with-file, batch + delete-all, refresh)	history	Angular component	Implemented	getHistory + getAbortedHisto deleteDownloadW
Cookies (status badge fresh/expiring/expired, age, MeTube online/offline, per-platform breakdown, upload, delete)	cookies	Angular component	Implemented	getCookieStatus uploadCookies + deleteCookies

Feature	Component	Category	Implementation	Wiring
Navbar (routerLinks Download/Queue/History/ Cookies; VPN / No-VPN / unknown profile badge; MeTube-unreachable badge)	navbar	Angular component	Implemented	getProfileStatu
Error boundary (renders loading / error+retry / empty / content states)	error-boundary	Angular component	Implemented	wrapped around lis components
Not-found (404 catch-all route)	not-found	Angular component	Implemented	** route in app.ro
Error interceptor (RxJS error surfacing, clears loading)	error- interceptor.service	Angular service	Implemented	app.config.ts pr
MeTube service (typed API: add/start/history/ delete/clear/retry/cookies/ profile/aborted/version/ poll; no any)	metube.service	Angular service	Implemented	HttpClient to /ap

6. TESTS / CHALLENGES (real test files reconciled)

Feature	Component	Category	Implementation	Wiring
Unit tests (pure-shell logic)	tests/test-unit.sh	Test suite	Implemented	run-tests.sh -p
		Test suite	Implemented	

Feature	Component	Category	Implementation	Wiring
Integration tests (containers up)	tests/test-integration.sh			run-tests.sh -p integration
Real-HTTP integration (no mocks, hits :9090/:8086/:8088)	tests/test-integration-realhttp.sh	Test suite	Implemented	services must be up
Scenario tests (lifecycle phases)	tests/test-scenarios.sh	Test suite	Implemented	run-tests.sh -p scenario
Error-phase tests	tests/test-errors.sh	Test suite	Implemented	run-tests.sh -p
Dashboard + landing integration	tests/test-dashboard.sh	Test suite	Implemented	services up
Dashboard operations (post-fix full automation)	tests/test-dashboard-operations.sh	Test suite	Implemented	services up
Aborted-history (incl. concurrent-posts no-500)	tests/test-aborted-history.sh	Test suite	Implemented	landing up
Cookie validator (_validate_cookie_file)	tests/test-cookie-validator.sh	Test suite	Implemented	landing app.py
		Test suite	Implemented	yt-dlp-cli / metube u

Feature	Component	Category	Implementation	Wiring
Media-services compatibility (anti-bluff documented-failure)	tests/test-media-services.sh			
Bulk operations	tests/test-bulk-operations.sh	Test suite	Implemented	services up
Add-all-platforms / add-download	tests/test-add-all-platforms.sh, tests/test-add-download.sh	Test suite	Implemented	metube up
VPN compose + smoke	tests/test-vpn-compose.sh, tests/test-vpn-smoke.sh	Test suite	Implemented	vpn profile
Chaos / resilience	tests/test-chaos.sh	Test suite	Implemented	services up
Constitution inheritance + meta-test	tests/test-constitution-inheritance.sh, tests/meta-test-constitution-inheritance.sh	Test suite	Implemented	repo tree
E2E (Playwright): dashboard, landing, cross-service, clear-history, delete-retry-cancel, cleanup-and-start	tests/e2e/	E2E suite	Implemented	run-e2e.sh + playwright.conf

Feature	Component	Category	Implementation	Wiring
Benchmark suite	tests/ benchmark/ run- benchmarks.sh	Benchmark suite	Implemented	services up
Cookie auth + youtube- download + validate-fix	tests/cookies/	Test suite	Implemented	metube + cookies
MeTube anti-bluff challenge bank (download- completes, queue-realtime, form-reenables)	tests/ challenges/	Challenge bank	Implemented	run-metube- challenges.sh + metube- challenges.json
Root HelixQA challenge scripts (api-contract, container-restart-resilience, download-completes, form- reenables, host/user OOM+suspend, landing- cookie-upload, memory- limits, no-vpn-direct, queue-lifecycle, queue- polling, retry-visible)	challenges/ scripts/	Challenge bank	Implemented	run_all_challeng
Challenges submodule (HelixQA challenge framework, p1-f06..f18 CLI-agent feature dirs)	Challenges/	Challenge framework (submodule)	Implemented (submodule)	Challenges/ catalo

7. MEDIA POST-PROCESSOR (media_postprocessor — shipped ytdlp-1.4.0)

Reconciled 2026-06-16 (Rev 4): these rows were marked Planned in Rev 3 but the dual-version pipeline shipped in tag ytdlp-1.4.0. Verified as FACT: real Python package media_postprocessor/ (config/jobs_db/media_probe/transcoder/watcher/worker/service), compose service media_postprocessor (container

media-postprocessor, MP_PORT=8089, MP_DB_PATH=/
 downloads/.media_postprocessor/jobs.db), nginx proxy dashboard/
 nginx.conf.template:116 /api/postprocess/ → media-
 postprocessor:8089, live endpoint returning real JSON, 6+ real webready-
 *.mp4 on disk.

Feature	Component	Category	Implementation	Wiring
Media post-processor sidecar (SQLite-WAL queue, watcher + worker, crash-safe resume, atomic output, ffprobe-validate, /postprocess/* API)	media_postprocessor	Service	Implemented (ytdlp-1.4.0)	Reachable: compose service media_postprocessor (containing media-postprocessor, MP_PORT=8089); dashboard nginx proxies /api/postprocess/media-postprocessor:8089 (nginx.conf.template:116)
Web-ready video transcode (webready-<base>.mp4, H.264/AAC, +faststart)	media_postprocessor	Pipeline feature	Implemented	transcoder.transcode_video → webready-<base>.mp4, ffprobe-validated (_validate_webready asserts h264 + faststart); watcher(+manager) + worker
MP3 audio derivation (<base>.mp3, 320 kbps CBR)	media_postprocessor	Pipeline feature	Implemented	transcoder.derive_mp3(<base>.mp3; worker invokes audio jobs
Post-process status API (/postprocess/status — healthy + per-state counts)	media_postprocessor	API feature	Implemented	media_postprocessor.serve_postprocess/status; nginx proxy api/postprocess/

Feature	Component	Category	Implementation	Wiring

8. PLANNED — not yet built (Implementation = Planned)

Confirmed ABSENT from the codebase, re-verified 2026-06-16 (Rev 4): no matching Angular component (find dashboard/src/app -iname '*postprocess*' -o -iname '*pipeline*' → none) and no user-facing download-resume feature — §11.4.6.

Feature	Component	Category	Implementation	Wiring	Tests Coverage	Val Ver
Dashboard pipeline-status UI (surface transcode/derive progress)	dashboard	Web UI feature (planned)	Planned	Not wired — no component exists	None yet	SK
Resume mechanism (resume interrupted/aborted downloads)	metube / media_postprocessor	Pipeline feature (planned)	Planned	Not wired (the postprocessor's crash-safe <i>job</i> resume is implemented; user-facing <i>download</i> resume is not)	None yet	SK

Reconciliation note (§11.4.153 / §11.4.118)

Every row above maps to real code (a named compose service, a Flask route in landing/app.py, an Angular component/service file under dashboard/src/app/, a root CLI script, a media_postprocessor/ module, or a test/challenge file) — except the §8 PLANNED rows, which are explicitly marked Planned/SKIP because the codebase contains no matching implementation (re-verified 2026-06-16). No code-present feature is omitted; no row lacks corresponding code. Per §11.4.6, no verdict claims a PASS without captured evidence. Per §11.4.153, exactly FIVE Video-Confirmation cells claim a real PASS — dashboard UI, landing 'Боџа' UI, MeTube UI, postprocess status API, and the ./status CLI — each backed by a ytdlp---prefixed captured artifact analyzed by Claude Opus 4.8 native multimodal, **all scope=render/transcript (none flow-confirmed)**; every other cell reads PENDING – not yet recorded.